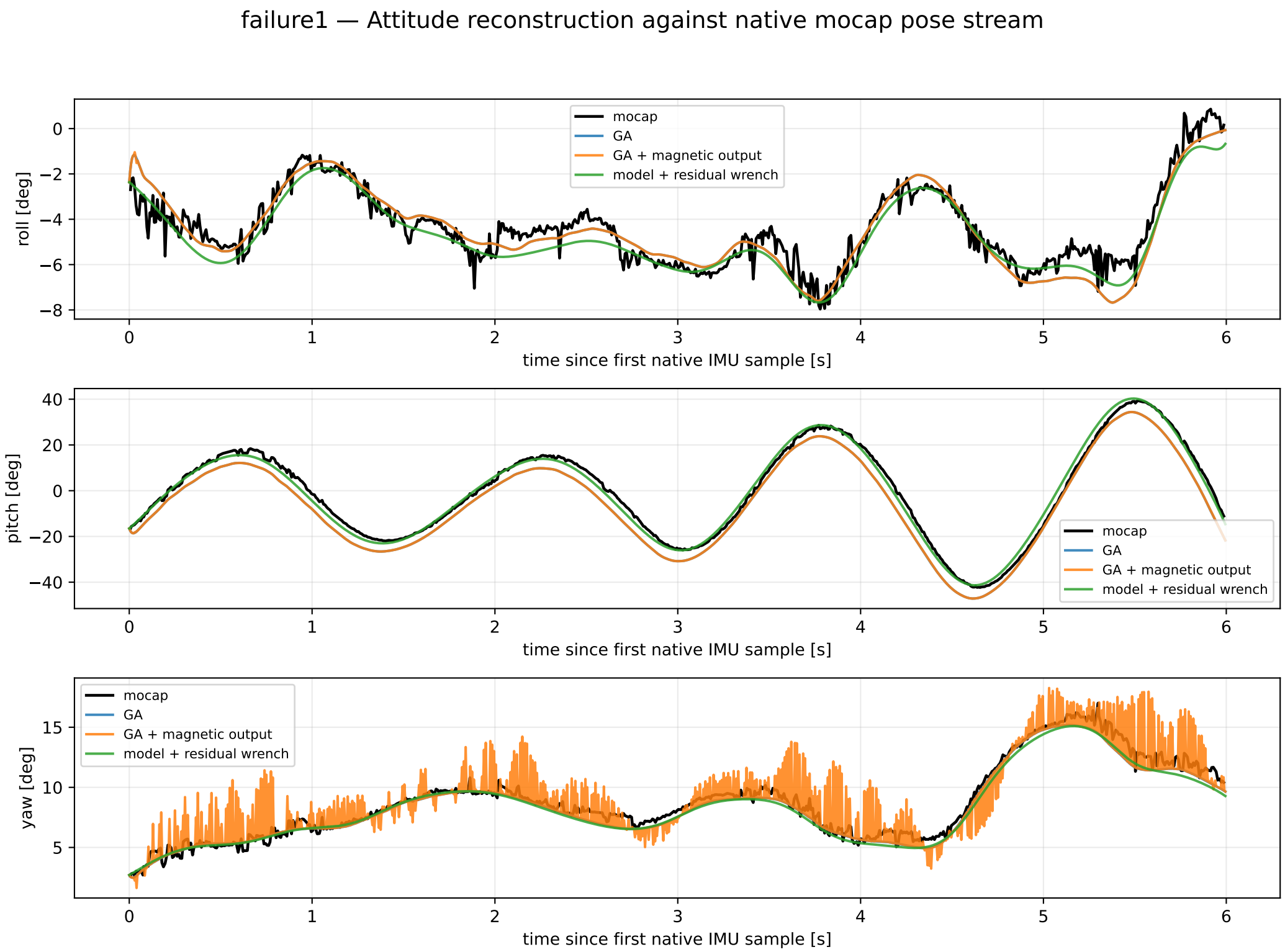
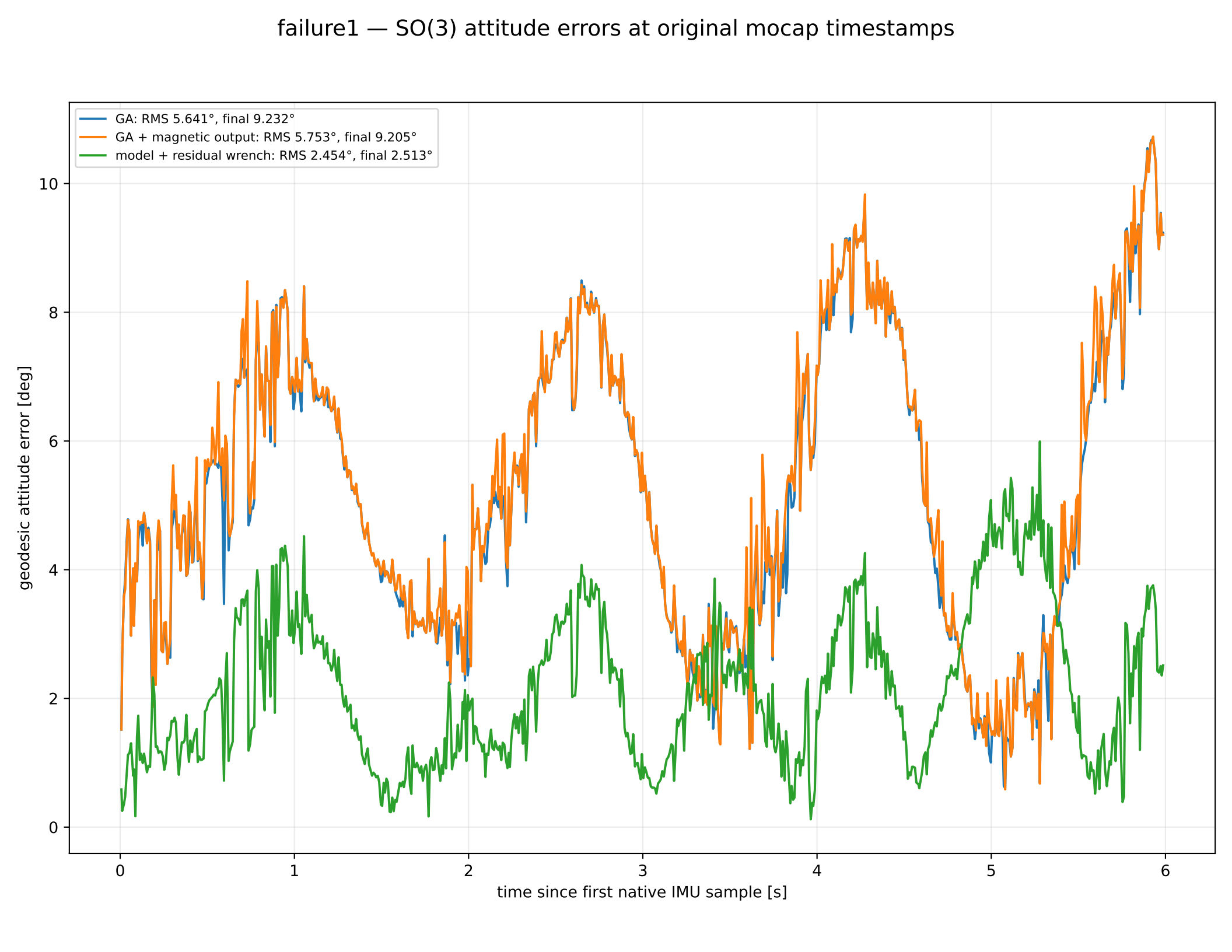
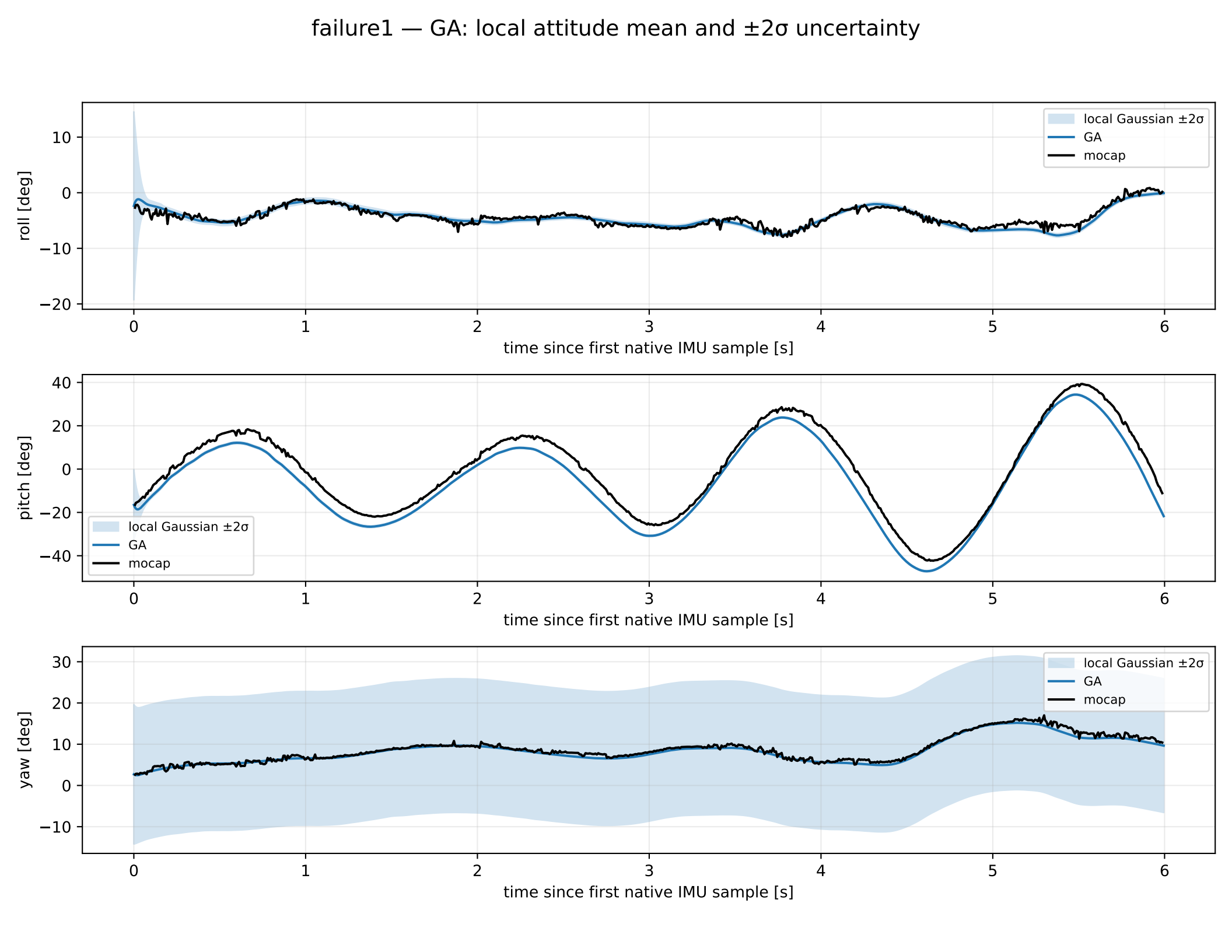
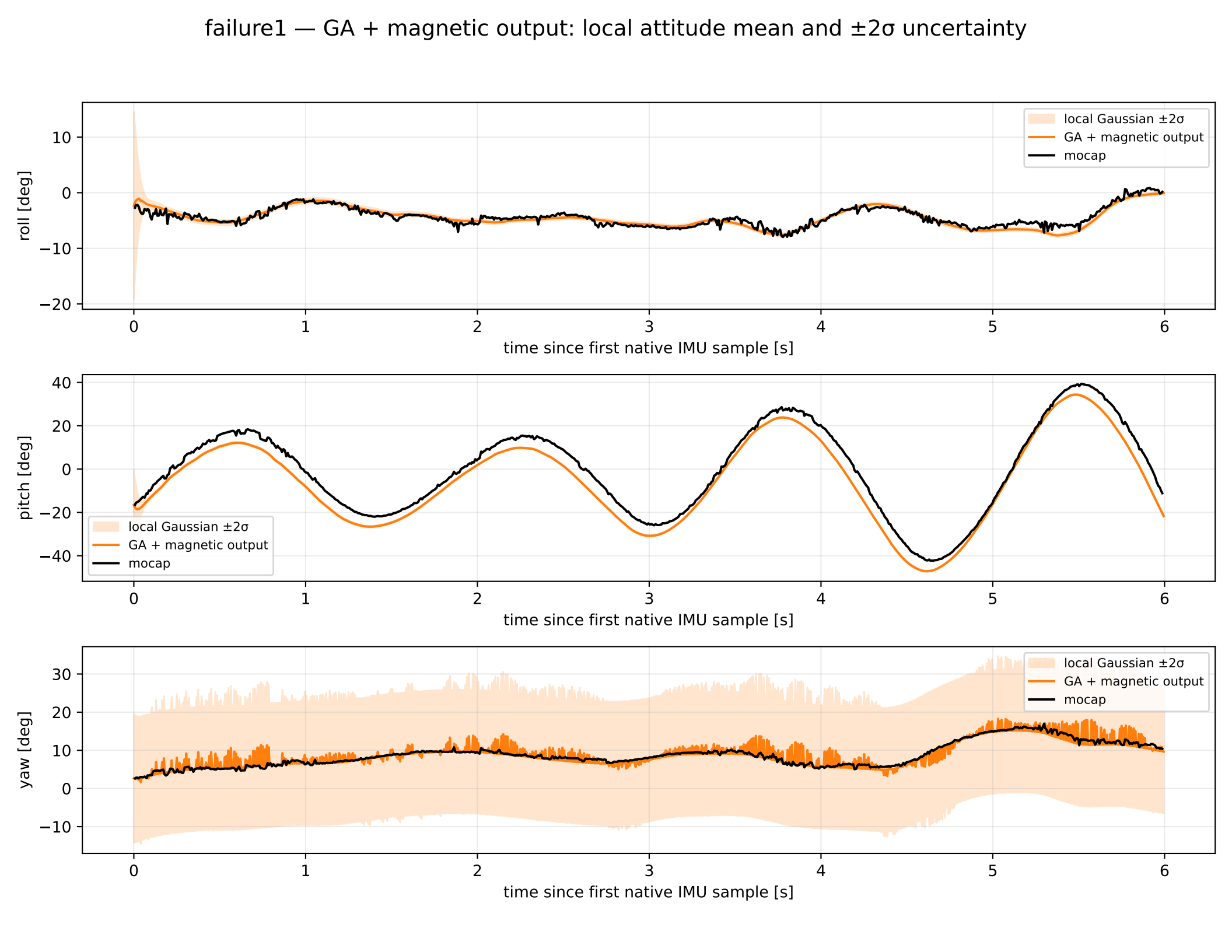
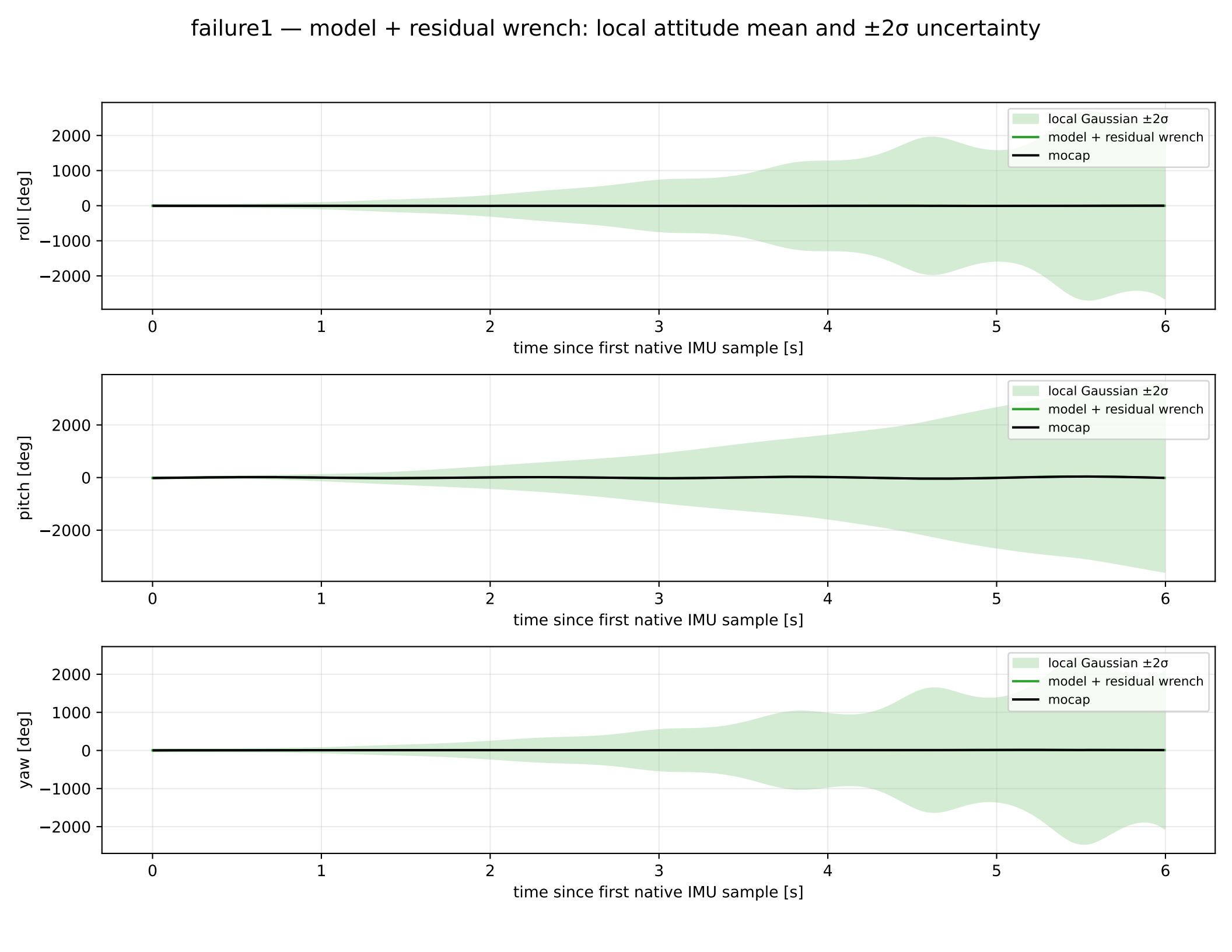

failure1 — Attitude reconstruction against native mocap pose stream
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
mocap
GA
GA + magnetic output
model + residual wrench

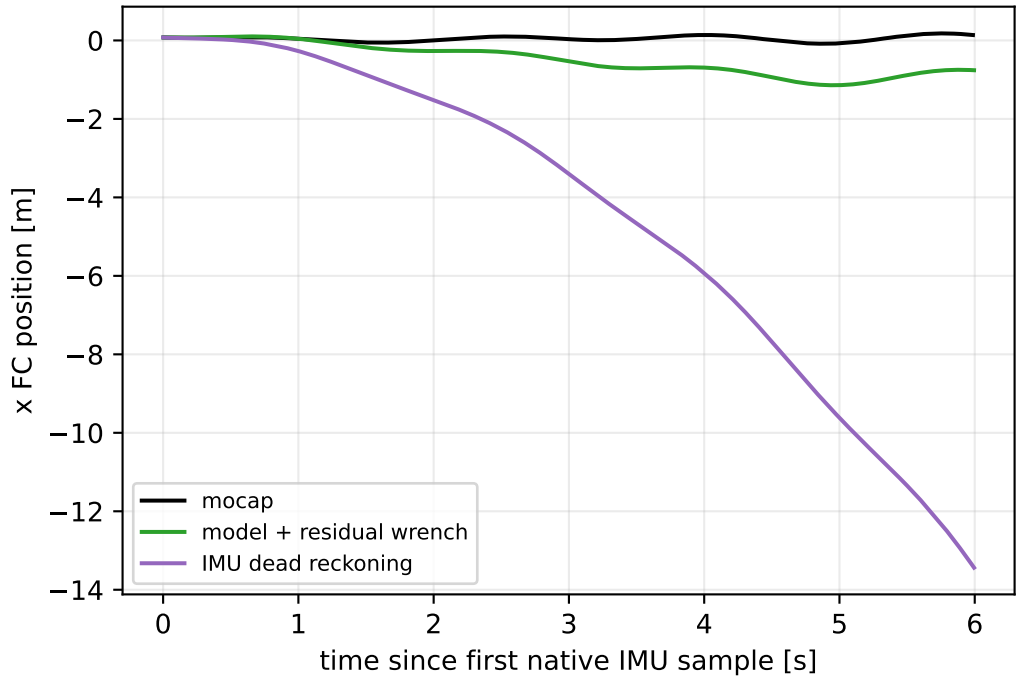

failure1 — SO(3) attitude errors at original mocap timestamps
GA: RMS 5.641°, final 9.232°
GA + magnetic output: RMS 5.753°, final 9.205°
model + residual wrench: RMS 2.454°, final 2.513°
geodesic attitude error [deg]
time since first native IMU sample [s]
0
2
4
6
8
10
0
1
2
3
4
5
6

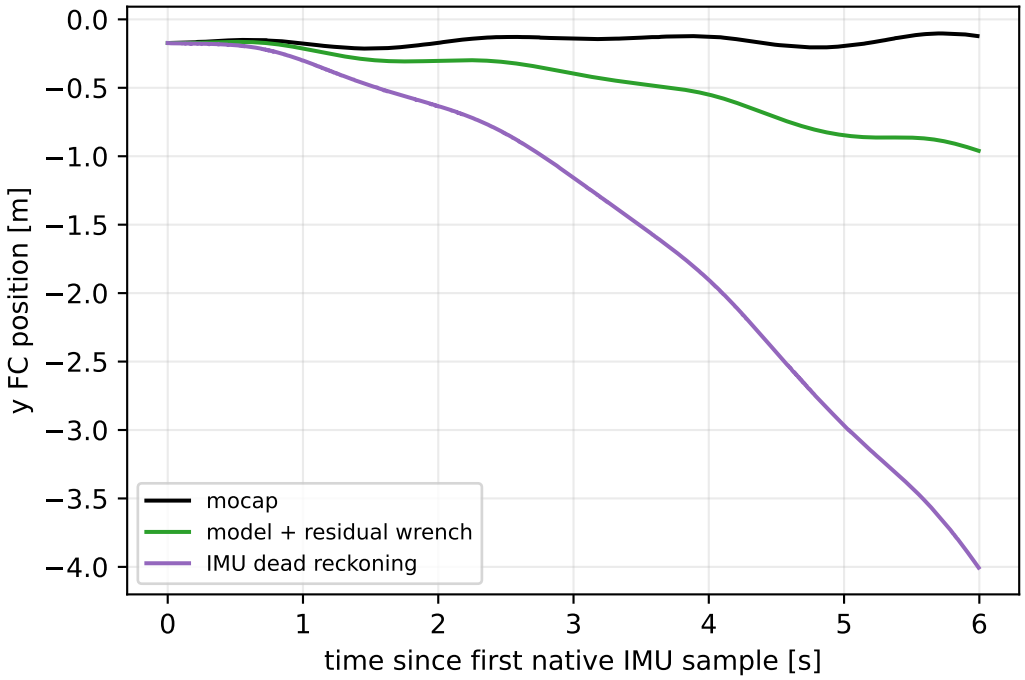

failure1 — GA: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA
mocap

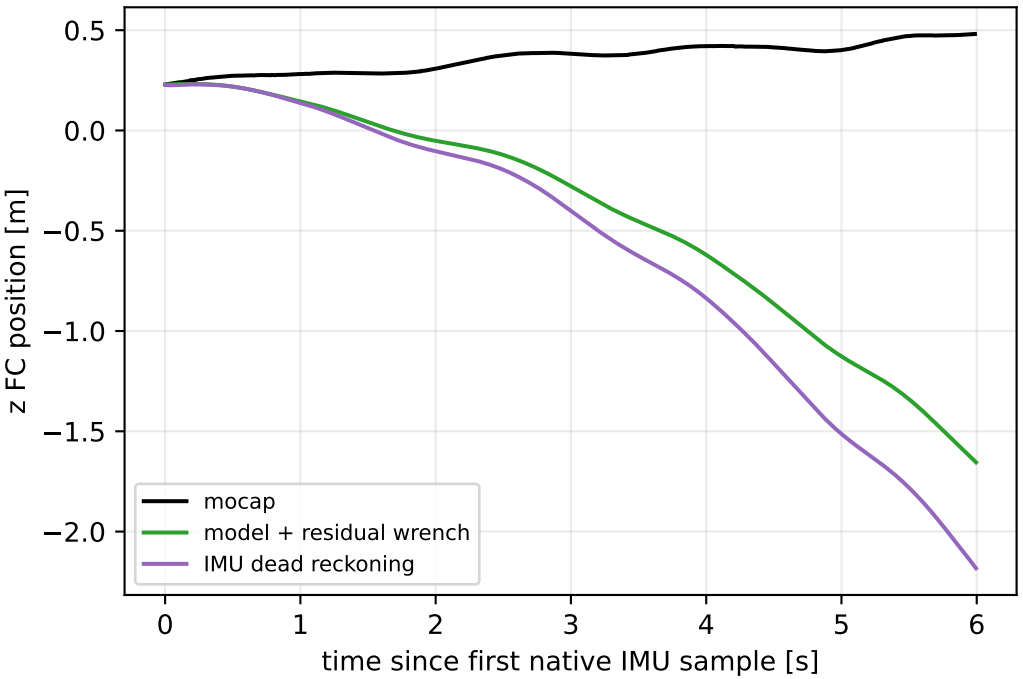

failure1 — GA + magnetic output: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA + magnetic output
mocap

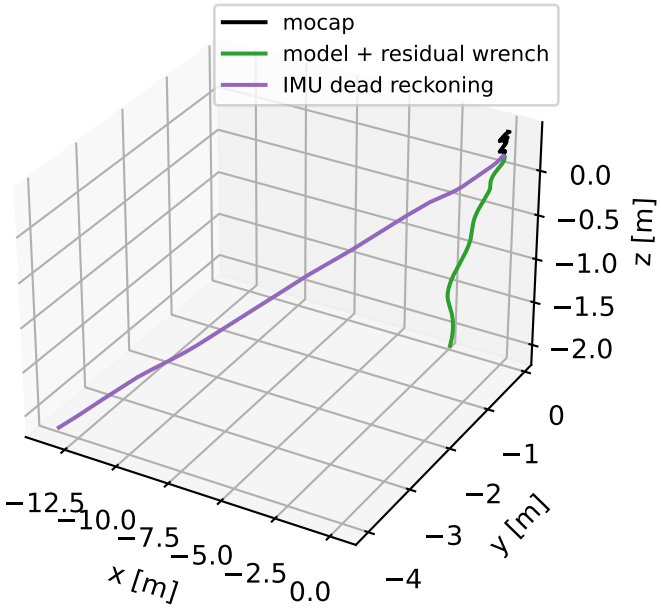

failure1 — model + residual wrench: local attitude mean and ±2σ uncertainty
local Gaussian ±2σ
model + residual wrench
mocap
roll [deg]
pitch [deg]
yaw [deg]
2000
1000
0
−1000
−2000
0
1
2
3
4
5
6
time since first native IMU sample [s]

failure1 — Position reconstruction (mean trajectories)



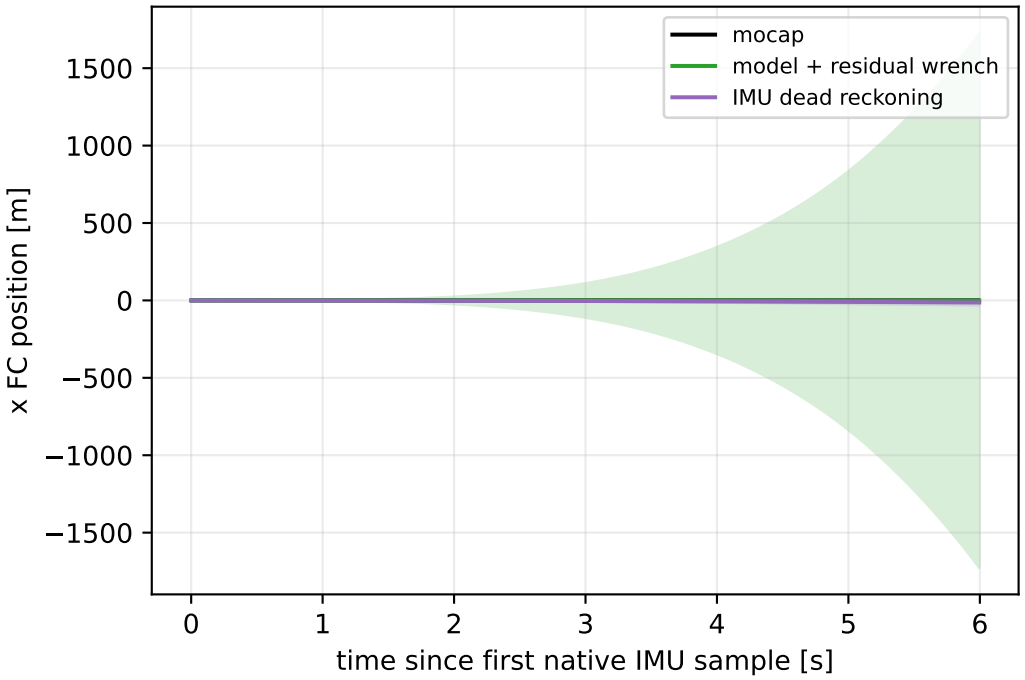


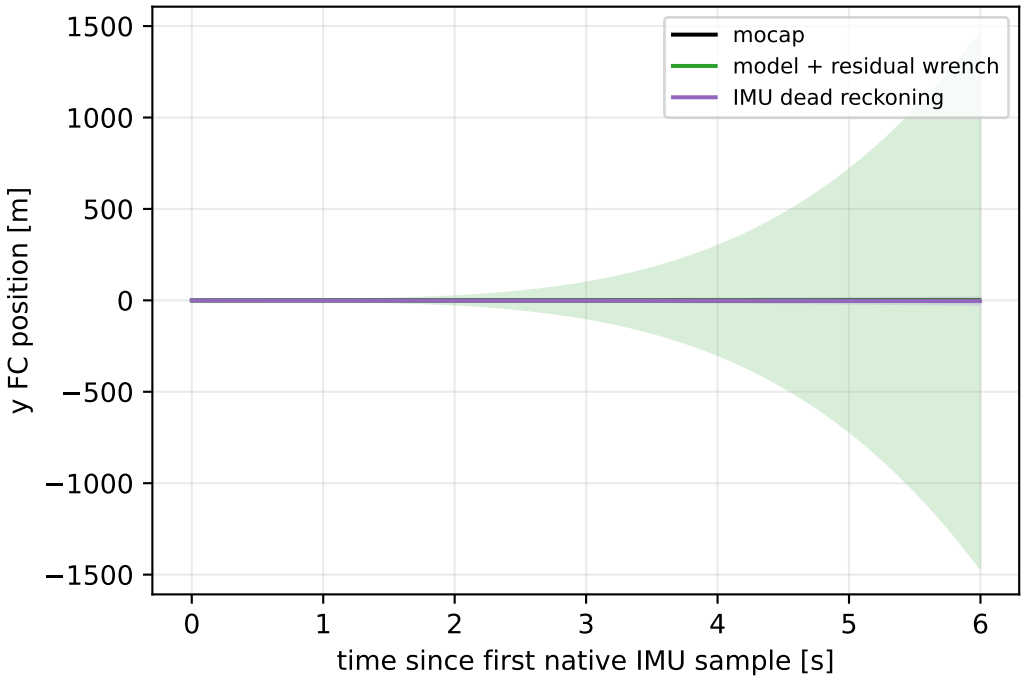


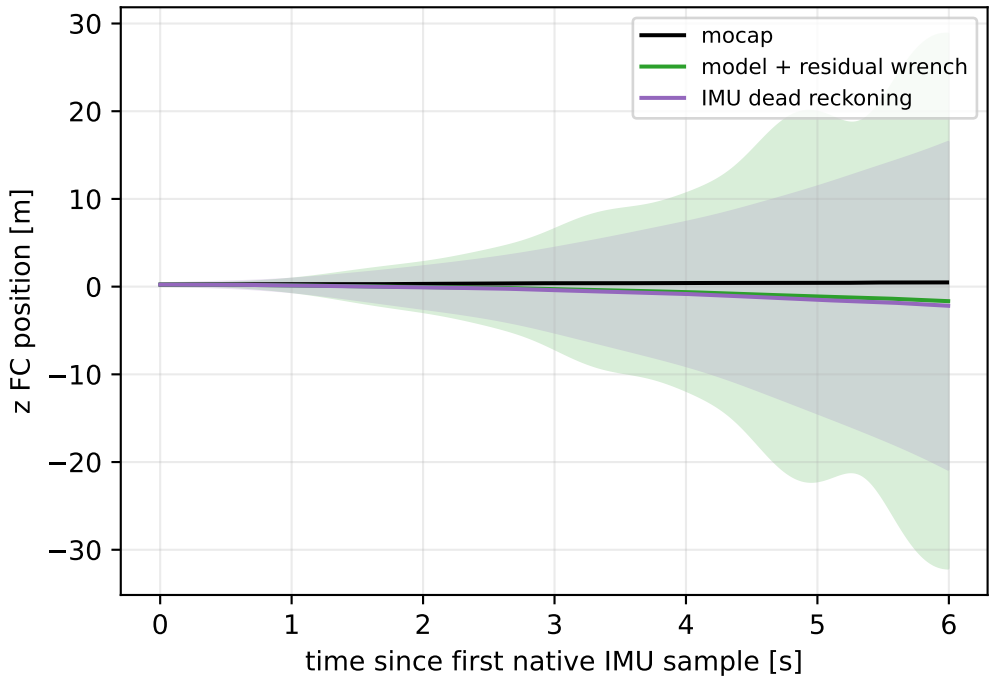


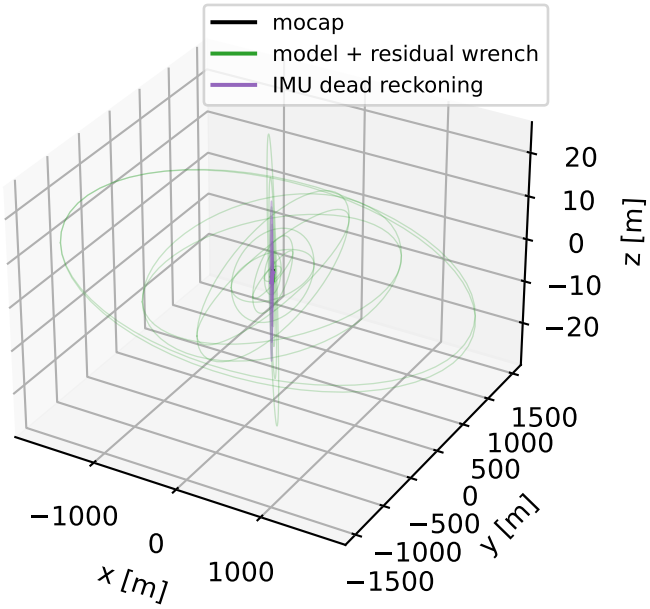
Residual-wrench model replay and IMU integration are not reset to later mocap.

failure1 — Position reconstruction with full propagated ±2σ spread









FC origin; local Gaussian linearization. Ellipsoids: deterministic 2σ factor-SVD wireframes, not samples.